De Bruijn's strip-narrowing theorem for the Riemann Ξ-function, formalized in Lean 4

Samuel Lavery

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Abstract

We report a complete, machine-checked proof in Lean 4 (Mathlib) of de Bruijn's 1950 theorem on the heat-flow deformations of the Riemann Ξ-function: writing $\Xi_\lambda(z) = \int_{\mathbb{R}} e^{\lambda u^2} K(u) e^{izu}\, du$ with $\Xi_0 = \Xi$, every Ξ_λ with $\lambda \geq \frac{1}{8}$ has only real zeros. In the normalization standard in the de Bruijn–Newman literature this is the bound $\Lambda \leq \frac{1}{2}$ on the de Bruijn–Newman constant. The development is self-contained — Mathlib contains no Laguerre–Pólya theory, no preservation of real-rootedness under locally uniform limits, no Hadamard product for ξ, and no de Bruijn–Newman material — so the Hadamard factorization of ξ, the polynomial strip-narrowing step, the passage from imaginary shifts to the heat semigroup, and a Hurwitz limit theorem are all built here. The result is 216 theorems over 14 modules and 3462 lines, with axiom footprint {`propext`, `Classical.choice`, `Quot.sound`} and no `sorry`. We also record a transversality theorem at the polynomial level — every real zero of the shift of a real-rooted polynomial is simple — together with an explicit account of why it does *not* descend to Ξ_λ. The Riemann Hypothesis is not proved and is not approached: the interval $\lambda \in (0, \frac{1}{8})$ remains, and closing it is equivalent to $\Lambda \leq 0$, which is RH. We close by reporting a numerical experiment locating where the $\frac{1}{8}$ is slack.

1 What is formalized

Let ξ be the completed Riemann zeta function and put

$$\Xi(z) \;=\; \xi\Big(\tfrac{1}{2} + iz\Big).$$

Then Ξ is entire, real on $\mathbb{R}$, and its zeros are the images of the nontrivial zeros of ζ: if $\rho = \beta + i\gamma$ is a nontrivial zero, the corresponding z is $z = \gamma - i(\beta - \frac{1}{2})$, so

$$\Xi(z) = 0 \implies |\operatorname{Im} z| = \Big|\beta - \tfrac{1}{2}\Big| < \tfrac{1}{2}, \tag{1}$$

because $0 < \beta < 1$. The Riemann Hypothesis is the assertion that every zero of Ξ is real.

Ξ has a classical Fourier representation $\Xi(z) = \int_{\mathbb{R}} K(u) e^{izu}\, du$ [6, §10.1] with an explicit even, positive, rapidly decaying kernel K built from the Jacobi theta function (§2). Deforming the kernel by a Gaussian gives the *flow*

$$\Xi_\lambda(z) \;=\; \int_{\mathbb{R}} e^{\lambda u^2} K(u)\, e^{izu}\, du, \qquad \Xi_0 = \Xi. \tag{2}$$

This is the backward heat flow: since $\partial_z^2 e^{izu} = -u^2 e^{izu}$, we have $\Xi_\lambda = e^{-\lambda\partial_z^2}\Xi$ formally, and the zeros of Ξ_λ migrate toward the real axis as λ increases. The object of this paper is the following theorem, proved by de Bruijn in 1950 ([1], Thm. 13) and here formalized. De Bruijn's statement is the strip contraction in the form $|\operatorname{Im} z| \le \big(\max(\Delta^2 - \lambda, 0)\big)^{1/2}$ for the kernel factor $e^{\lambda t^2/2}$, which with the factor $e^{\lambda u^2}$ used here reads $\big(\max(\Delta^2 - 2\lambda, 0)\big)^{1/2}$ — the law appearing as Proposition 6 below.

Theorem 1 (`flowFrontier_one_eighth`)**.** *For every λ with $\frac{1}{8} \le \lambda \le 1$, every zero of Ξ_λ is real.*

In Lean, with `flowStage` the function (2):

```
def FlowFrontier (lam₀ : ℝ) : Prop :=
  ∀lam : ℝ, lam₀ ≤lam →lam ≤1 →∀z : ℂ, flowStage lam z = 0 →z.im = 0

theorem flowFrontier_one_eighth : FlowFrontier (1 / 8)
```

The upper cut $\lambda \le 1$ is a convenience of the ambient development, not a restriction of the method; the narrowing argument is monotone in λ and `FlowFrontier.mono` transports the conclusion to every $\lambda_0 \ge \frac{1}{8}$.

The mechanism has three parts, and each is elementary in isolation:

(i) a *polynomial* step: the map $p \mapsto p(\cdot + ih) + p(\cdot - ih)$ carries a real polynomial whose zeros lie in $(\operatorname{Im} w)^2 \le \sigma$ to one whose zeros lie in $(\operatorname{Im} w)^2 \le \max(\sigma - h^2, 0)$ (§3–§4);

(ii) an *approximation* step: Ξ is a locally uniform limit of real polynomials with all zeros in $(\operatorname{Im} w)^2 \le \frac{1}{4}$, supplied by the Hadamard product (§7);

(iii) a *limit* step: iterating (i) with N shifts of size h where $Nh^2 = 2\lambda$ realizes the Gaussian $e^{\lambda u^2}$ in the limit, and real-rootedness survives locally uniform limits by Hurwitz (§6–§9).

2 Normalization

The de Bruijn–Newman literature is not uniform in its scaling, and the numerical value of the threshold depends on it, so we fix conventions explicitly.

Let $\psi(x) = \sum_{n\ge1} e^{-\pi n^2 x}$ and set $\Psi(u) = \psi(e^{2u})$. The functional equation $\psi(1/x) = \sqrt{x}\,\psi(x) + \frac{\sqrt{x}-1}{2}$ becomes

$$\Psi(u) = e^{-u}\Psi(-u) + \tfrac{e^{-u}-1}{2}, \tag{3}$$

which is compiled as `Psi_reflect` (from Mathlib's `evenKernel_functional_equation` at $a = 0$; in the formalization Ψ is the restriction to $\mathbb{R}$ of `PsiC`$(w) = \frac{1}{2}(\vartheta_2(0, ie^{2w}) - 1)$, holomorphic on $|\operatorname{Im} w| < \pi/4$). The kernel is

$$K(u) \;=\; e^{u/2}\big(\Psi''(u) + \Psi'(u)\big),$$

and differentiating (3) twice gives $\Psi''(u) + \Psi'(u) = e^{-u}(\Psi''(-u) + \Psi'(-u))$, hence $K(-u) = K(u)$: the kernel is even (`Kker_even`). That $\int_{\mathbb{R}} K(u)e^{izu}du = \Xi(z)$ is `xiSection_eq_Ktrans`, proved by analytic continuation from the half-plane $\operatorname{Im} z < -\frac{1}{2}$ where the integral converges absolutely term by term.

With this u, the strip bound (1) reads $\Delta = \frac{1}{2}$, and Theorem 1 gives real-rootedness at $\lambda \geq \frac{1}{8}$. The de Bruijn–Newman constant Λ is defined in the literature through the family H_t normalized so that $H_0(z) = \frac{1}{8}\Xi(z/2)$; halving the z-variable doubles the half-width to $\Delta_{\mathrm{std}} = 1$ and divides the Gaussian exponent by four, so

$$\lambda \;=\; t/4. \tag{4}$$

Relation (4) is not ours: Ki–Kim–Lee state it explicitly, defining the de Bruijn–Newman constant as $\Lambda = 4\lambda(0)$ with $\lambda(0)$ the infimum of λ for which Ξ_λ has only real zeros [3]. Theorem 1 is therefore exactly de Bruijn's published bound $\Lambda \leq \frac{1}{2}$ [1]. Two further consistency checks: de Bruijn's $t = \frac{1}{2}$ maps to $\lambda = \frac{1}{8}$, and the Polymath 15 bound $t \leq 0.22$ [4] maps to $\lambda \leq 0.055$, correctly strictly better than $\frac{1}{8}$. A reader who prefers the H_t normalization should read every λ below as $t/4$.

3 The elementary core: one identity

Everything in part (i) rests on a single algebraic identity, which we state first because it is the whole mechanism and it is two lines.

For real a, b let $Q_{a,b}(z) = (z-a)^2 + b^2$, the real quadratic with roots $a \pm ib$.

Lemma 2 (`pairFactor_normSq_shift`)**.** *For all real a, b, h and all $z \in \mathbb{C}$,*

$$\left|Q_{a,b}(z+ih)\right|^2 - \left|Q_{a,b}(z-ih)\right|^2 \;=\; 8\,(\operatorname{Im} z)\,h\,\Big((\operatorname{Re} z - a)^2 + (\operatorname{Im} z)^2 + h^2 - b^2\Big).$$

Proof. Expand both sides in $\operatorname{Re} z$, $\operatorname{Im} z$ and compare; in Lean this is `simp` on the real and imaginary parts followed by `ring`. □

The consequence is immediate and is the engine of the whole paper.

Corollary 3 (`pairFactor_norm_shift_lt`)**.** *If $\operatorname{Im} z > 0$, $h > 0$ and $b^2 < (\operatorname{Im} z)^2 + h^2$, then $|Q_{a,b}(z-ih)| < |Q_{a,b}(z+ih)|$.*

In words: *a conjugate pair of roots that is not too high up sees the upward shift as strictly larger than the downward one*, uniformly in the real part. A real polynomial p factors over $\mathbb{C}$ into such quadratics together with real linear factors, and a real linear factor $z - a$ is the case $b = 0$ of $Q_{a,b}$ after squaring; taking the product of strict inequalities over all roots (`polynomial_norm_shift_lt`) gives

Proposition 4 (`polynomial_shift_zero_strip`)**.** *Let $p \in \mathbb{R}[X]$, $p \neq 0$, $h > 0$, and suppose every root w of p satisfies $(\operatorname{Im} w)^2 \leq \Delta^2$. If $p(z+ih) + p(z-ih) = 0$ then*

$$\operatorname{Im} z = 0 \qquad \text{or} \qquad (\operatorname{Im} z)^2 + h^2 \leq \Delta^2.$$

Proof. If $\operatorname{Im} z > 0$ and $(\operatorname{Im} z)^2 + h^2 > \Delta^2$ then Corollary 3 applies to every root, so $\|p(z-ih)\| < \|p(z+ih)\|$; but $p(z+ih) = -p(z-ih)$ forces equality of norms, a contradiction. The case $\operatorname{Im} z < 0$ follows by conjugating, using that p has real coefficients (`Polynomial.aeval_conj`), which exchanges the two shifted arguments. No simplicity assumption on the roots is used: multiplicities are carried by the multiset of roots throughout. □

We emphasise what Proposition 4 costs: a single `ring` identity, a product of strict inequalities, and the triangle inequality. There is no potential theory and no Laguerre–Pólya theory in it.

4 The strip class and the shift

Definition 5 (`StripClass`)**.** For $\sigma \in \mathbb{R}$ and $f : \mathbb{C} \to \mathbb{C}$, say f lies in the *strip class at* σ if there are nonzero real polynomials P_n with every root of every P_n satisfying $(\operatorname{Im} w)^2 \le \sigma$, such that $P_n \to f$ locally uniformly on $\mathbb{C}$.

```
def StripClass (σ : ℝ) (f : ℂ→ℂ) : Prop :=
  ∃P : ℕ→ℝ[X], (∀ n, P n ≠0) ∧(∀ n (w : ℂ), aeval w (P n) = 0 →w.im ^ 2 ≤σ) ∧
    TendstoLocallyUniformly (fun n z => aeval z (P n)) f atTop
```

Write $S_h f(z) = f(z + ih) + f(z - ih)$ for the imaginary shift operator.

Proposition 6 (`StripClass.shift`, `StripClass.shiftIter`)**.** *Let $\sigma \ge 0$, $h > 0$. If f lies in the strip class at σ then $S_h f$ lies in the strip class at $\max(\sigma - h^2, 0)$; consequently $S_h^N f$ lies in the strip class at $\max(\sigma - Nh^2, 0)$.*

Proof. The polynomial witnesses transport: S_h applied to a real polynomial is again a real polynomial, because the complex polynomial $p(X + ih) + p(X - ih)$ has real coefficients (the odd-order Hasse derivatives cancel; `shiftPolyC_coeff_real`), and its root strip is controlled by Proposition 4. Local uniform convergence is preserved because precomposition with $z \mapsto z \pm ih$ preserves it and sums of locally uniformly convergent sequences converge locally uniformly. Nonvanishing of the shifted witness is `shiftPoly_ne_zero`. The iterate is an induction on N, whose only content is the arithmetic $\max(\max(\sigma - Nh^2, 0) - h^2, 0) = \max(\sigma - (N{+}1)h^2, 0)$. □

5 Transversality: the shift crosses, it never grazes

The narrowing step of §4 controls *where* the zeros of $S_h f$ are. A separate and equally elementary argument controls *how* they are met: at the polynomial level the shift's real zeros are always simple. This is not needed for Theorem 1, and we record it because it is the exact point at which multiplicity enters the construction, and because it is sharp (Remark 10).

The observation that starts it: p has real coefficients, so for real x and real h, $p(x - ih) = \overline{p(x + ih)}$, whence on the axis

$$(S_h p)(x) = 2 \operatorname{Re} p(x + ih), \qquad (S_h p)'(x) = 2 \operatorname{Re} p'(x + ih). \tag{5}$$

So a *double* zero of $S_h p$ at x says exactly that both $p(w)$ and $p'(w)$ are purely imaginary at $w = x + ih$ — and then $\overline{p(w)}\, p'(w)$ is real. It never is:

Lemma 7 (`im_conj_mul_deriv_neg`)**.** *Let $p \in \mathbb{R}[X]$ have positive degree and only real zeros, and let $\operatorname{Im} w > 0$. Then*

$$\operatorname{Im}\big(\overline{p(w)}\, p'(w)\big) < 0.$$

The reason is the classical one — $\overline{p}p'/|p|^2 = \sum_k 1/(w-r_k)$ and each term has imaginary part $-\operatorname{Im} w/|w-r_k|^2 < 0$ — but the formalization does not construct that sum. It peels one real root at a time: writing $p = (X-r)q$ with r real and q again real-rooted,

$$\operatorname{Im}(\overline{p(w)}p'(w)) \;=\; -\,|q(w)|^2\,\operatorname{Im} w \;+\; |w-r|^2\,\operatorname{Im}(\overline{q(w)}q'(w)), \tag{6}$$

in which the first term is strictly negative — a real-rooted q cannot vanish off the axis, so $|q(w)|^2 > 0$ — and the second is ≤ 0 by induction on the degree. The strict sign is produced by the peel itself, which is why no sum and no partial-fraction decomposition is required.

Theorem 8 (`shiftPoly_zero_simple`)**.** *Let $p \in \mathbb{R}[X]$ have positive degree and only real zeros, and let $h > 0$. Then every real zero of $S_h p$ is simple.*

Proof. At a zero, (5) gives $\operatorname{Re} p(w) = 0$; if it were not simple we would also have $\operatorname{Re} p'(w) = 0$, making $\overline{p(w)}p'(w)$ real and contradicting Lemma 7. □

The statement is carried to the compiled shift polynomial by `shiftPoly_derivative` : $\partial(S_h p) = S_h(\partial p)$ — the shift commutes with differentiation — so Theorem 8 is a statement about `derivative (shiftPoly h p)` and not merely about the shifted evaluations.

Remark 9 (why this does *not* descend to Ξ_λ)**.** Theorem 8 is a statement about each approximating polynomial, and simplicity is not a closed condition: Hurwitz (§8) carries real-rootedness to a locally uniform limit but not simplicity, since a sequence of simple zeros may coalesce. One might hope to recover it from a uniform lower bound on the zero gaps of the approximants, but no such bound is available at this level: the approximants have degree $\to \infty$, and the heat flow applied to z^n produces the Hermite polynomial, whose n roots spread over a range $\sim \sqrt{n}$ so that the minimum gap degrades. Any proof of simplicity for Ξ_λ must therefore draw on the limit's own zero density rather than on the polynomial step. We do not prove it.

Remark 10 (the threshold is sharp for simplicity)**.** Simplicity cannot hold at $\lambda = \frac{1}{8}$ itself. The isolated pair $p(z) = z^2 + \frac{1}{4}$ flows to $p_t(z) = z^2 + \frac{1}{4} - 2t$, which at $t = \frac{1}{8}$ is exactly z^2 — a genuine double zero — and for $t > \frac{1}{8}$ has the two distinct real roots $\pm\sqrt{2t - \frac{1}{4}}$. So the correct statement to aim at is simplicity on the *open* range $\lambda > \frac{1}{8}$, matching the classical assertion that the zeros of Ξ_λ are real for $\lambda \ge \frac{1}{8}$ and simple for $\lambda > \frac{1}{8}$. Numerically the separation is immediate rather than gradual: applying the flow to real-rooted polynomials with multiple roots (z^2, z^3, z^4, $(z^2)(z-1)^2$, $(z+1)^2 z(z-1)^2$) gives all-real all-simple roots already at $t = 10^{-8}$, with minimum gap growing like $2\sqrt{2t}$.

6 From shifts to the heat flow

On the Fourier side the shift is a multiplication operator:

$$S_h\Big(\int K(u)e^{izu}du\Big) = \int K(u)(e^{-hu} + e^{hu})e^{izu}du = \int 2\cosh(hu)\,K(u)\,e^{izu}\,du.$$

Iterating, $S_h^N \Xi(z) = \int \big(2\cosh(hu)\big)^N K(u)e^{izu}du$; this is `shiftOp_iterate_Ktrans`. Now choose

$$h_n = \sqrt{\frac{2\lambda}{n+1}}, \qquad N = n+1, \qquad \text{so that} \qquad N h_n^2 = 2\lambda \;\text{ for every } n,$$

and normalize by $2^{-(n+1)}$. Since $\cosh(h_n u)^{n+1}$ is squeezed between $\big(1+\frac{\lambda u^2}{n+1}\big)^{n+1}$ and $e^{\lambda u^2}$ — the lower bound from $1 + r^2/2 \le \cosh r$, the upper from $\cosh r \le e^{|r|}$ — dominated convergence gives:

Proposition 11 (`tendstoLocallyUniformly_coshApprox`)**.** *For $0 \le \lambda \le 1$, the normalized stages $z \mapsto \int \cosh(h_n u)^{n+1} K(u) e^{izu} du$ converge to Ξ_λ locally uniformly on $\mathbb{C}$.*

The dominating function is uniform on each horizontal strip $|\operatorname{Im} z| \le A$, which is what "locally uniformly on $\mathbb{C}$" requires; the compact-set formulation is obtained from `tendstoLocallyUniformly_iff_forall_isCompact`.

Two nonvanishing facts are needed and are proved separately: no weighted transform $\int w(u) K(u) e^{izu} du$ with w continuous, positive and of at most exponential growth is identically zero (`coshStage_ne_zero`, `flowStage_ne_zero`). These follow from Fourier inversion together with $\xi(\frac{1}{2}) \ne 0$. They are exactly the hypothesis "f is not identically zero" of the Hurwitz step, and omitting them would make the argument vacuous rather than wrong, which is why we flag them.

7 Ξ lies in the strip class at $\frac{1}{4}$

This is the arithmetic input, and the only place where anything about ζ beyond (1) is used.

Proposition 12 (`xiSection_stripClass`)**.** *Ξ lies in the strip class at $\sigma = \frac{1}{4}$.*

Construction of the witnesses. Start from the Hadamard product, compiled as `riemannXi_hadamard_factorization`:

$$\xi(s) = e^{As+B} \prod_{\rho} \Big(1 - \frac{s}{\rho}\Big)^{m_\rho} e^{m_\rho s/\rho},$$

the product over nontrivial zeros with multiplicity. Restrict to the cofinal finite sets $G_N = \{\rho : |\operatorname{Im}\rho| \le N\}$, which are closed under $\rho \mapsto 1-\rho$ and $\rho \mapsto \bar\rho$ by the functional equation and the reflection $\overline{\xi(\bar s)} = \xi(s)$. Substituting $s = \frac{1}{2} + iz$ and writing $w_\rho = -i(\rho - \frac{1}{2})$,

$$e^{As+B} \prod_{\rho \in G_N} \Big((1-\tfrac{s}{\rho})e^{s/\rho}\Big)^{m_\rho} \;=\; \kappa_N\, e^{iz(A+c_N)}\, Q_N(z), \qquad Q_N(z) = \prod_{\rho \in G_N} \Big(1 - \frac{z}{w_\rho}\Big)^{m_\rho},$$

with $c_N = \sum_{G_N} m_\rho/\rho$. Because G_N is closed under the two involutions, the zero multiset of Q_N is closed under conjugation, so $Q_N \in \mathbb{R}[X]$; and w_ρ satisfies $|\operatorname{Im} w_\rho| = |\operatorname{Re}\rho - \frac{1}{2}| < \frac{1}{2}$ by (1), so every root of Q_N lies in $(\operatorname{Im} w)^2 \le \frac{1}{4}$. The paired sums converge, $c_N \to -A$ (`reflection_const_eq_zero`), so the exponential factor tends to 1 and $\kappa_N \to \Xi(0) \ne 0$. Hence $\Xi(0)\, Q_N \to \Xi$ locally uniformly, and after rescaling these are the required witnesses. □

The cancellation $c_N \to -A$ is the one delicate point: individually the sums $\sum 1/\rho$ do not converge absolutely, and it is the pairing $\rho \leftrightarrow 1-\rho$ inside each G_N that makes c_N converge. This is why the exhaustion must be by symmetric sets and not by, say, the first N zeros in some order.

8 Real zeros survive locally uniform limits

Proposition 13 (`Hurwitz.im_eq_zero_of_tendstoLocallyUniformly`)**.** *Let F_n be entire with only real zeros, $F_n \to f$ locally uniformly on $\mathbb{C}$, and f not identically zero. Then f has only real zeros.*

The formalization proves this through a nonvanishing form of Hurwitz's theorem on an open preconnected set, and it is worth recording the route because it avoids winding numbers entirely.

Lemma 14 (`Hurwitz.ne_zero_of_tendstoLocallyUniformlyOn`)**.** *Let $U \subseteq \mathbb{C}$ be open and preconnected, F_n holomorphic on U with F_n nowhere zero on U, $F_n \to f$ locally uniformly on U, and $f \not\equiv 0$ on U. Then f is nowhere zero on U.*

Proof. Suppose $f(z_0) = 0$, $z_0 \in U$. Since $f \not\equiv 0$ on the preconnected U, the identity theorem gives a punctured neighbourhood of z_0 on which $f \neq 0$; choose $r > 0$ with $\overline{B}(z_0, r) \subseteq U$ and $f \neq 0$ on the sphere $|z - z_0| = r$, and let $m = \min_{|z-z_0|=r} \|f(z)\| > 0$ (attained, the sphere being compact). Local uniform convergence is uniform on $\overline{B}(z_0, r)$, so for some n we have $\|F_n - f\| < m/2$ there, whence

$$\|F_n(z)\| > m/2 \ \text{ on the sphere}, \qquad \|F_n(z_0)\| < m/2 \ \text{ at the centre}.$$

But F_n is nowhere zero on the disc, so $1/F_n$ is holomorphic there, and the maximum modulus principle bounds $\|1/F_n(z_0)\|$ by its maximum on the boundary, i.e. $\le 2/m$ — that is, $\|F_n(z_0)\| \ge m/2$, contradicting the second display. □

Proof of Proposition 13. Apply Lemma 14 to the two open half-planes $U_\pm = \{\pm \operatorname{Im} w > 0\}$. Each is convex, hence preconnected; each F_n is nowhere zero on $U_\pm$ precisely because its zeros are all real; and $f \not\equiv 0$ on $U_\pm$ because $f \not\equiv 0$ on $\mathbb{C}$ and f is entire, so vanishing on an open subset would force vanishing identically. Hence f has no zero off the real axis. □

Specialised to the strip class this gives `StripClass.im_eq_zero_of_zero`: a member of the strip class at $\sigma = 0$ which is not identically zero has only real zeros.

9 Assembly

Proof of Theorem 1. Fix λ with $\frac{1}{8} \le \lambda \le 1$ and let $n \in \mathbb{N}$. By Proposition 12, Ξ lies in the strip class at $\frac{1}{4}$. Apply Proposition 6 with $h = h_n$ and $N = n + 1$: the iterate $S_{h_n}^{n+1}\Xi$ lies in the strip class at

$$\max\Big(\tfrac{1}{4} - (n{+}1)h_n^2,\, 0\Big) = \max\Big(\tfrac{1}{4} - 2\lambda,\, 0\Big) = 0, \qquad \text{since } \lambda \ge \tfrac{1}{8}.$$

By `shiftOp_iterate_Ktrans` that iterate is the cosh stage, and by Proposition 13 (applied within the strip class, using `coshStage_ne_zero`) every zero of the n-th normalized stage is real. Now let $n \to \infty$: by Proposition 11 the stages converge to Ξ_λ locally uniformly, and $\Xi_\lambda \not\equiv 0$ by `flowStage_ne_zero`, so Proposition 13 applies once more and every zero of Ξ_λ is real. □

Hurwitz is used twice, at two different levels — once to pass from polynomial witnesses to each fixed stage, once to pass from the stages to the flow. The threshold $\frac{1}{8}$ enters at exactly one place: the requirement $2\lambda \geq \sigma = \frac{1}{4}$.

Corollary 15 (`flowCascade_realRooted_of_le_seven`)**.** *In the cascade $\lambda_n = 1/(n+1)$, the stages $n \leq 7$ have only real zeros, at every height.*

10 The formalization

The development is Lean 4 against Mathlib [7] at the revision pinned in `lake-manifest.json` (`v4.33.0-rc2`).

module	lines	thms	content
`XiKernelTheta`	123	16	Ψ holomorphic on a strip
`XiKernelEven`	122	11	$K(-u) = K(u)$ from the theta reflection
`XiKernelTransform`	269	22	$\int K(u)e^{izu}du = \Xi(z)$
`XiHadamardFactorization`	484	13	the Hadamard product for ξ
`XiProductMultPartialFraction`	277	6	multiplicity bookkeeping, $\sum m_\rho/\rho$
`XiStripApproximants`	542	43	Prop. 12: Ξ in the class at $\frac{1}{4}$
`DeBruijnStripStep`	230	13	Lem. 2, Prop. 4
`HurwitzLimit`	139	2	Prop. 13
`StripShiftClass`	166	14	`StripClass`, Prop. 6
`FiberShiftSimple`	226	7	Lem. 7, Thm. 8
`XiFlowCascade`	366	30	Ξ_λ, the cascade λ_n
`FlowCoshStages`	354	26	Prop. 11, nonvanishing
`FlowFrontier`	69	5	`FlowFrontier`, monotonicity
`DeBruijnFlowFrontier`	95	8	Theorem 1
total	3462	216	

Axiom audit. Every named theorem in the chain, checked with `#print axioms`, depends on exactly {`propext`, `Classical.choice`, `Quot.sound`} — the three axioms of Lean's standard classical foundation — and nothing else. There is no `sorry` and no bespoke `axiom` in the development or anywhere in the repository containing it: a full `lake build` over 10451 jobs reports zero errors and zero declarations using `sorry`.

What Mathlib supplied and what it did not. Mathlib supplied the analytic infrastructure: holomorphy, locally uniform convergence, the maximum modulus principle, the identity theorem (`eqOn_zero_of_preconnected_of_eventuallyEq_zero`), dominated convergence, Fourier integrals, the Jacobi theta function with its functional equation (`evenKernel_functional_equation`), polynomial algebra over $\mathbb{R}$ and $\mathbb{C}$ including `aroots` with multiplicity, and `completedRiemannZeta` with its functional equation. Mathlib did *not* supply, and this development therefore builds from scratch: the Hadamard product for ξ; any notion of a real-rooted or Laguerre–Pólya class; preservation of real-rootedness under locally uniform limits; and any de Bruijn–Newman material whatsoever. A search of the pinned revision for

`Laguerre`, `Pólya`, `realRooted` and `deBruijn` returns nothing relevant; `Mathlib/Analysis/-Complex/Hadamard.lean` is the three-lines theorem, not the factorization.

11 The remaining interval

Let `FlowFrontier` λ_0 abbreviate "every Ξ_λ with $\lambda_0 \le \lambda \le 1$ has only real zeros". The following is also compiled:

```
theorem rh_of_flowFrontier (h : ∀lam₀ : ℝ, 0 < lam₀ →FlowFrontier lam₀) :
    RiemannHypothesis
```

Its hypothesis quantifies over *every* positive λ_0; the proof instantiates at $\lambda_n = 1/(n+1) \to 0$. Theorem 1 supplies $\lambda_0 \ge \frac{1}{8}$. The gap is therefore the bounded interval

$$\lambda \in \left(0, \tfrac{1}{8}\right),$$

which under (4) is the statement $\Lambda \le 0$, and by Newman's conjecture [2], settled by Rodgers–Tao ($\Lambda \ge 0$) [5], this is equivalent to the Riemann Hypothesis. Nothing in this paper bears on it.

It is worth recording precisely *where* the constant $\frac{1}{8}$ comes from, because it is not a feature of Ξ. Proposition 4 is exactly tight for a single conjugate pair and nothing else: for $p(z) = z^2 + y_0^2$ one has $p'' = 2$, so under the flow $p_t(z) = z^2 + y_0^2 - 2t$, whose roots reach the real axis exactly at $t = y_0^2/2$; with $y_0 = \frac{1}{2}$ this is $\frac{1}{8}$. The input $\sigma = \frac{1}{4}$ of Proposition 12 is the bare critical strip (1): no zero density, no zero count, no arithmetic beyond $0 < \beta < 1$ enters Theorem 1 at all.

Remark 16 (a numerical experiment, not a theorem)**.** The flow acts exactly on polynomials, $p_t = \sum_j (-t)^j p^{(2j)}/j!$, and for a conjugate pair at $x \pm iy$ in a field of further zeros x_k the imaginary part obeys

$$\partial_t y \;=\; -\frac{1}{y} \;-\; 2y \sum_k \frac{1}{(x - x_k)^2 + y^2},$$

in which *both* terms are negative when the x_k are real: neighbours accelerate the descent, so $t^* \le y_0^2/2$ should be slack. We measured this. With $y_0 = \frac{1}{2}$ and m real zeros at spacing $s = 1/\rho$, the first time t^* at which all roots are real satisfies (control: $m = 0$ reproduces $t^* = \frac{1}{8}$ to 8×10^{-23})

ρ		1	2	4
$t^*/(1/8)$	$(m = 40)$	0.471	0.279	0.156

and at $\rho = 4$ the value 0.0195 agrees to 2% with the linear law $y_0/(2\pi\rho) = 0.0199$ that replaces the quadratic $y_0^2/2$ once the density is felt. Since the zeros of Ξ have density $\rho = \log(T/2\pi)/2\pi$ at height T, the corresponding heuristic threshold is $y_0/\log(T/2\pi) \to 0$. Replacing all neighbours by conjugate pairs at the maximal height $y_1 = y_0 = \frac{1}{2}$ — the configuration best able to push the pair *away* from the axis, and the relevant one since the neighbours of a hypothetical off-line zero are not known to be real — the ratios become 0.680, 0.479, 0.329: still decreasing in ρ. De Bruijn's bound held as a control throughout ($t^* \le \frac{1}{8}$ in every configuration tested).

We draw no conclusion about Ξ from this. What it does identify is the point at which a sharpening would have to enter: a *local, density-aware* refinement of Proposition 6, in which m roots in a window of width w about x buy a narrowing near x strictly better than Nh^2, by an amount increasing in m/w. That is a statement about polynomials.

12 What is not claimed

The theorem proved here is Theorem 1 and nothing more. In particular:

- The Riemann Hypothesis is not proved, and no step of this development is a step toward it. The distance between Theorem 1 and RH is the interval $\lambda \in (0, \frac{1}{8})$, and closing that interval *is* RH; it is not a gap in formalization or in plumbing.
- Theorem 1 is de Bruijn's 1950 bound [1] and is weaker than the published state of the art. Ki–Kim–Lee sharpened it to the strict inequality $\Lambda < \frac{1}{2}$ [3], and Polymath 15 gave $\Lambda \le 0.22$ [4], i.e. $\lambda \le 0.055$, roughly a factor 2.3 better; neither is formalized here, and formalizing Polymath 15 would shrink the open interval to $(0, 0.055)$ without closing it.
- The novelty claimed is formalization, not mathematics. The mathematical content of §3–§9 is de Bruijn's, and where the argument is arranged differently from his — the strip class as an explicit predicate, the cosh-power staging, the symmetric exhaustion G_N — that is a convenience of formalization, not a new idea.
- Simplicity of the zeros of Ξ_λ is *not* proved. Theorem 8 is a statement about the approximating polynomials only; Remark 9 explains why it does not descend through the limit, and Remark 10 why any such statement must be confined to the open range $\lambda > \frac{1}{8}$. A fortiori nothing here bears on the simplicity of the zeros of ζ, which would be a statement at $\lambda = 0$.
- Remarks 10 and 16 report numerical experiments on polynomials, with pre-registered decision rules and positive controls. They are not theorems and are not used anywhere above.

Availability

The development builds with `lake build` against the pinned Mathlib. The two results stated above are

Theorem 1	`CriticalLinePhasor.XiKernel.flowFrontier_one_eighth` in `RequestProject/DeBruijnFlowFrontier.lean`
Theorem 8	`CriticalLinePhasor.DeBruijn.shiftPoly_zero_simple` in `RequestProject/FiberShiftSimple.lean`

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